

KITALE, Kenya

WMO: 636610

Lat: 0.972N

Lon: 34.959E

Elev: 1850

StdP: 80.99

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.8	11.4	1.7	5.3	24.2	3.2	6.0	23.4	7.7	21.1	6.1	20.3	0.8	90	0.307

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.9	29.9	15.2	28.9	15.1	28.0	15.3	19.2	24.5	18.7	23.9	18.4	23.5	3.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	15.8	21.2	17.1	15.4	20.6	16.8	15.1	20.2	63.7	24.6	62.1	24.1	60.7	23.5	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.3	5.2	4.4	DB	8.1	31.7	2.3	0.8	6.4	32.3	5.1	32.8	3.8	33.3	2.1	33.9	(4)
(5)			WB	6.4	21.9	1.6	1.9	5.2	23.2	4.3	24.3	3.4	25.4	2.3	26.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	19.4	20.0	20.8	20.9	20.4	19.7	18.9	18.4	18.3	18.7	19.1	19.0	19.3	(6)
(7)		DBStd	1.43	1.28	1.31	1.58	1.34	1.01	0.96	0.96	1.09	1.07	1.04	0.91	1.10	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	56	2	1	1	1	2	5	11	13	8	4	4	3	(9)
(10)		CDD10.0	3446	310	302	337	312	299	267	259	258	261	283	271	287	(10)
(11)		CDD18.3	460	54	69	80	63	43	22	12	12	18	29	24	33	(11)
(12)		CDH23.3	3475	604	716	702	370	182	84	54	59	158	155	127	264	(12)
(13)		CDH26.7	572	106	179	186	63	10	2	0	1	7	3	2	13	(13)
(14)	Wind	WSAvg	1.5	1.5	1.8	1.7	1.5	1.3	1.2	1.3	1.4	1.6	1.5	1.4	1.5	(14)
(15)	Precipitation	PrecAvg	1237	24	57	89	167	168	99	128	148	88	93	84	35	(15)
(16)		PrecMax	1839	115	182	191	411	325	235	239	281	194	227	365	132	(16)
(17)		PrecMin	881	0	0	5	45	39	12	10	3	3	16	6	3	(17)
(18)		PrecStd	197	30	55	54	71	69	51	63	64	53	57	76	30	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.1	31.0	31.2	30.2	27.9	26.6	25.8	26.1	27.8	27.1	26.9	27.9	(19)
MCWB			14.9	14.7	15.0	16.1	18.1	17.9	17.7	17.9	16.2	16.8	17.3	15.7	(20)	
2%		DB	28.8	29.7	30.0	28.4	26.3	25.2	24.7	24.9	26.2	26.0	25.7	26.8	(21)	
		MCWB	14.9	14.8	15.1	16.1	18.0	17.7	17.6	17.5	16.6	16.9	16.9	15.7	(22)	
5%		DB	27.9	28.8	28.8	27.0	25.2	24.2	23.9	23.9	25.2	25.1	24.9	25.9	(23)	
		MCWB	14.9	14.7	15.1	16.5	17.8	17.5	17.4	17.3	16.8	16.9	16.8	15.7	(24)	
10%		DB	26.8	27.8	27.3	25.6	24.2	23.2	22.9	22.9	24.0	24.1	23.9	24.9	(25)	
		MCWB	15.0	14.7	15.5	16.9	17.7	17.3	17.1	17.1	16.9	16.9	16.7	15.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.3	18.3	18.7	19.6	19.7	19.4	19.0	19.0	19.1	19.0	19.1	18.8	(27)
MCDWB			24.5	24.5	24.3	25.1	25.3	24.5	24.2	24.5	24.9	24.2	24.1	23.6	(28)	
2%		WB	17.3	17.4	17.9	18.8	19.1	18.6	18.2	18.2	18.3	18.3	18.3	18.1	(29)	
		MCDWB	23.6	23.7	23.8	23.7	24.4	23.7	23.3	23.4	23.8	23.5	23.2	23.0	(30)	
5%		WB	16.7	16.9	17.4	18.2	18.5	18.1	17.7	17.7	17.6	17.8	17.8	17.4	(31)	
		MCDWB	22.9	23.0	23.1	23.1	23.6	23.0	22.5	22.6	22.8	22.9	22.6	22.3	(32)	
10%		WB	16.2	16.3	17.0	17.8	18.1	17.5	17.2	17.2	17.1	17.3	17.3	16.9	(33)	
		MCDWB	22.6	22.7	22.5	22.3	22.8	22.1	21.8	21.8	22.0	22.2	21.8	21.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.9	14.9	13.4	11.3	10.4	10.6	10.5	10.7	11.8	11.2	11.0	12.6	(35)
MCDBR			16.5	16.6	16.0	13.6	12.2	11.7	11.8	12.0	13.7	12.6	12.5	14.3	(36)	
MCWBR			5.6	5.1	4.5	4.1	4.9	5.2	5.3	5.4	5.7	5.0	5.1	5.5	(37)	
5% WB		MCDWB	13.9	14.0	12.2	10.8	10.9	10.9	11.0	11.3	12.0	11.2	10.9	11.7	(38)	
		MCWBR	5.4	5.0	4.3	4.2	4.8	5.2	5.4	5.4	5.5	5.0	5.1	5.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.334	0.354	0.362	0.352	0.363	0.418	0.431	0.405	0.369	0.357	0.343	0.337	(40)	
taud		2.528	2.465	2.463	2.542	2.503	2.303	2.251	2.333	2.435	2.476	2.532	2.531	(41)		
Ebn at Noon		994	976	959	949	915	851	845	887	938	960	977	986	(42)		
Edh at Noon		108	117	117	106	106	127	135	128	119	115	107	107	(43)		
(44)	All-Sky Solar Radiation	RadAvg	6.27	6.55	6.39	5.91	5.80	5.41	5.22	5.51	6.14	5.94	5.69	5.94	(44)	
(45)		RadStd	0.43	0.36	0.42	0.37	0.20	0.29	0.26	0.27	0.24	0.22	0.31	0.38	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page